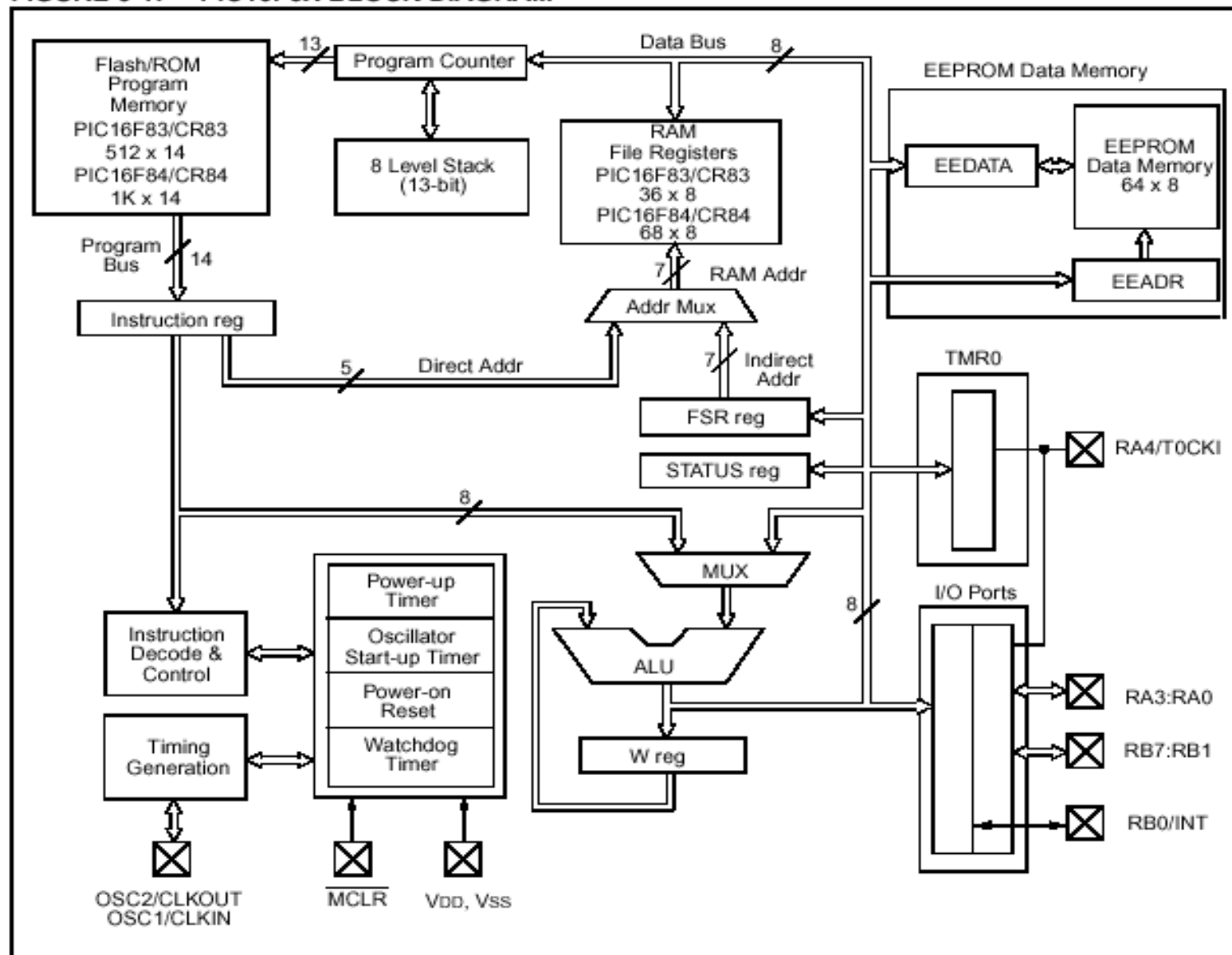


PIC-Microcontroller

FIGURE 3-1: PIC16F8X BLOCK DIAGRAM



- File Select Register-FSR
- timer0 Register- tmR0
- WREG stands for Working Register.
- Asynchronous Communication (e.g., UART/RS-232, RS-485)
 - (No shared clock; both sides must agree on a baud rate).
- Synchronous Communication (e.g., SPI, I2C, USART in Sync mode)
 - (Requires a shared clock signal (CLK) between transmitter and receiver)
 - Baud rate in PIC microcontrollers defines the speed of serial data transmission (UART) in bits per second (bps)

PIC Microcontrollers: PIC

- Microcontrollers are popular in low-end applications. They use RISC (Reduced Instruction Set Computer) processors. RISC processors have fewer instructions and addressing modes. They execute one instruction per internal clock cycle. This leads to less complex hardware and reduced silicon area and power consumption.
- The architecture includes separate program memory and data memory. The program counter generates addresses for the program memory.
- A hardware stack is used to handle calls and interrupts, saving the program counter value for return.
- The instruction register fetches instructions based on the program counter.
- Instructions are decoded to produce control and timing signals.
- The oscillator or clock provides timing, including the watchdog timer, which is crucial for embedded systems.
- The ALU performs arithmetic and logic functions.
- RAM, also known as the register file, stores data and status bits.
- The bus structure directs data flow, and registers maintain arithmetic and status information.

PIC Architecture

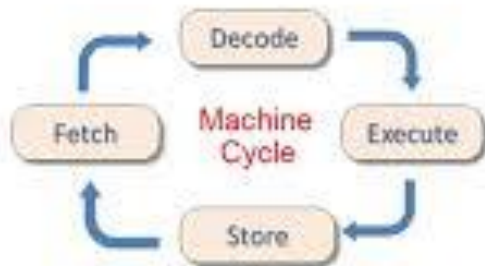
PICs are "RISC"

Reduced Instruction Set Computer
(RISC)

Few instructions (usually < 50)

Only a few addressing modes

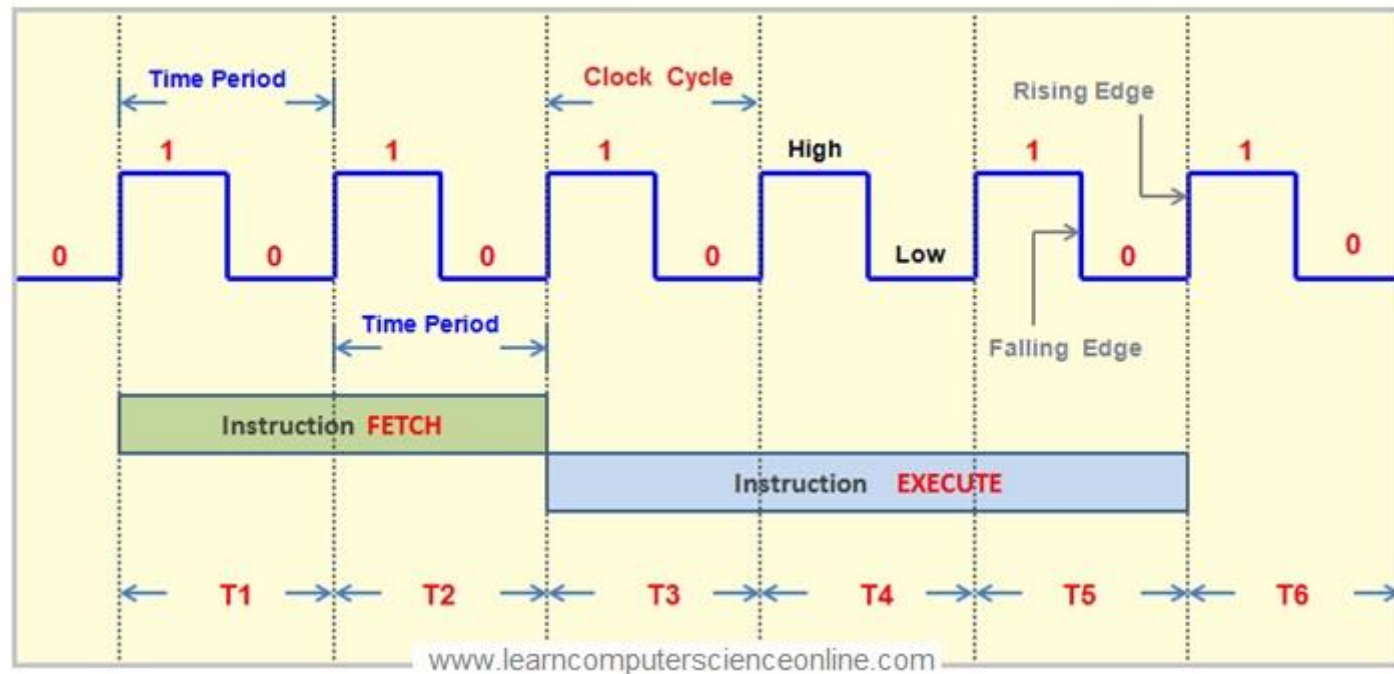
Executes 1 instruction in 1 internal clock
cycle (T_{cyc})



What Is CPU Machine Cycle ?

www.learncomputerscienceonline.com

The System Clock Speed And Instruction Cycle



The PIC Family: Packages

PICs come in a huge variety of packages:

Examples:

8 pin	:	12C50x (12bit) and 12C67x (14bit)
18pin	:	16C5X (12bit), 16Cxxx (14bit)
28pin	:	16C5X (12bit), 16Cxxx (14bit)
40pin	:	16Cxxx (14bit), 17C4x (16bit)
44 - 68pin	:	16Cxxx (14bit), 17C4x / 17Cxxx (16bit)

1.CPU (Central Processing Unit):

PIC microcontroller's CPU consists of:

- Arithmetic logic unit (ALU)
- Memory unit (MU)
- Control unit (CU)
- Accumulator

ALU is used to perform arithmetic and logical operations. After the instructions have been processed, memory is employed to store them. The internal and external peripherals attached to the CPU are controlled by the control unit, and the results are stored in the accumulator. It supports normally RISC architecture.

Reduced Instruction Set Computer, or RISC, is a microprocessor design that uses a limited, highly optimized set of instructions rather than the more specialised set of instructions found in other architectures.

2.MEMORY ORGANIZATION:

PIC micro controller memory module consists of mainly 3 types of memories

•PROGRAM MEMORY:

It contains the programme that was written and burned into the microcontroller. Program Counter sequentially executes commands contained in the programme memory. The Pic microcontroller has an 8K words x 14 bit Flash programme memory that can be erased and reprogrammed electrically. We erase an old programme and write a new one every time we burn a programme into the micro.

•DATA MEMORY:

It is a form of RAM that is used to temporarily store data in its registers. Banks are used to organize RAM memory. Each bank has a limit (128 bytes). Depending on the microcontroller, the number of banks may vary. There are only two banks on the PIC16F84. Special Function Registers (SFR) and General Purpose Registers (GPR) are both found in banks (GPR). Special Function Registers are kept in the bottom locations of each bank, whereas General Purpose Registers are kept in the upper places.

2. MEMORY ORGANIZATION:

- General Purpose Registers (GPR):**

These registers have no particular purpose. These are used for general multiplication, addition, and subtraction, with the results being stored in other registers. The data in these registers is easily accessible by the CPU.

- Special Function Registers (SFR):**

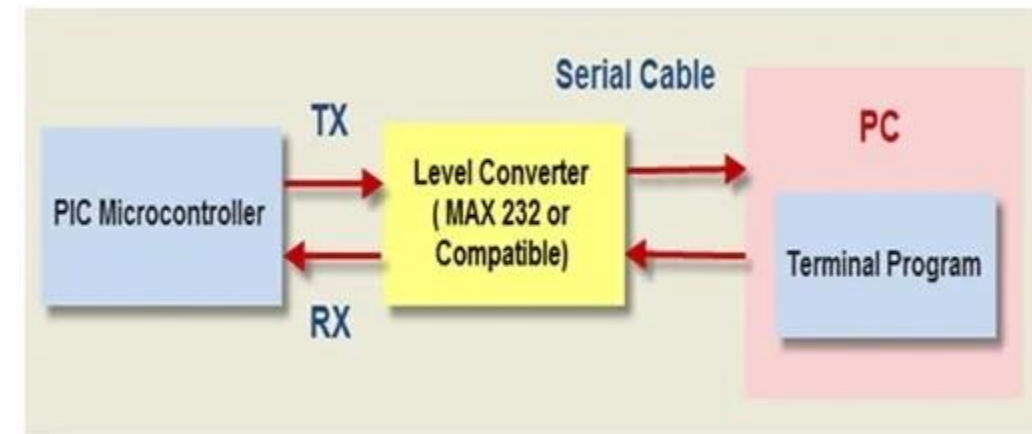
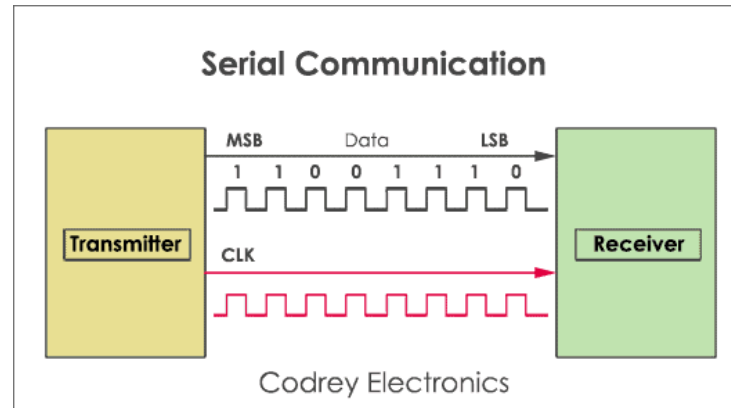
These registers are used for special purposes and they cannot be used as normal registers. Their function is set at the time of manufacturing. They perform the function assigned to them and user cannot change the function of SFR. Three important SFRs for programming are:

- STATUS register: It changes the bank
- PORT registers: It assigns logic values 0 or 1 to the ports
- TRIS registers: It functions as a data direction register to configure pins as input (1) or output (0).

- DATA EEPROM:**

This memory permits variables to be stored as a result of burning a programme. During regular functioning, it is readable and writable (over the full VDD range). The register file does not directly map this memory. The SFRs are used to address it in a roundabout way. This memory is read and written with the help of six SFRs (EECON1, EECON2, EEDATA, EEDATH, EEADR, EEADRH).

3. SERIAL COMMUNICATION:



The transfer of one bit of data at time consecutively over a communication channel is called Serial Communication. There are **three protocols** of serial communication:

- **USART:** It stands for **Universal synchronous and Asynchronous Receiver and Transmitter** which provides a serial communication in two devices. In this protocol data is transmitted and received bit by bit through a single wire according to the clock pulses. To send and receive data serially the PIC microcontroller has two pins TXD and RXD.
- **SPI Protocol:** (**SPI**) is an **acronym for Serial Peripheral Interface**. It is used to communicate data **between PIC microcontrollers and other devices such as sensors, shift registers, and SD cards**. The PIC microcontroller supports three-wire SPI communications between two devices using the same clock source. The SPI protocol is capable of handling more data than the USART protocol.
- The I2C protocol, which stands for Inter Integrated Circuit, is used to connect low-speed devices such as microcontrollers, EEPROMs, and A/D converters. The PIC microcontroller supports two-wire interface (I2C) communication between two devices that can function as both **Master and Slave**.

3.SERIAL COMMUNICATION:

Feature 	UART	I2C	SPI
Type	Asynchronous	Synchronous	Synchronous
Wires	2 (TX, RX)	2 (SDA, SCL)	4+ (MISO, MOSI, SCK, CS)
Speed	Slow to Medium	Moderate	Fastest
Duplex	Full-Duplex	Half-Duplex	Full-Duplex
Distance	Long	Short (On-board)	Short (On-board)
Best For	Debugging, GPS	Sensors, EEPROM	LCDs, SD Cards

Serial Data (SDA) and Serial Clock (SCL)

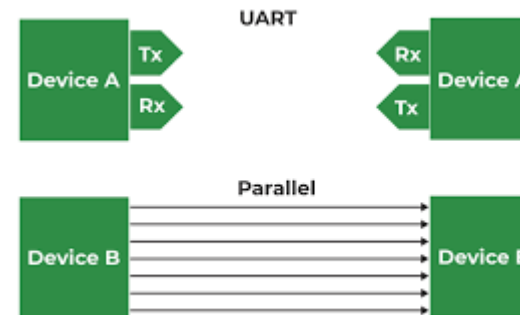
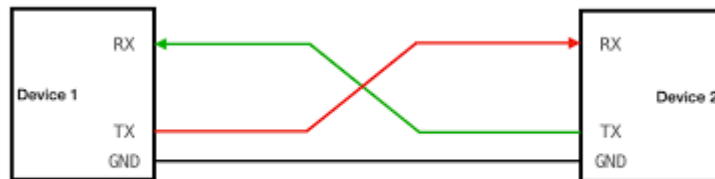
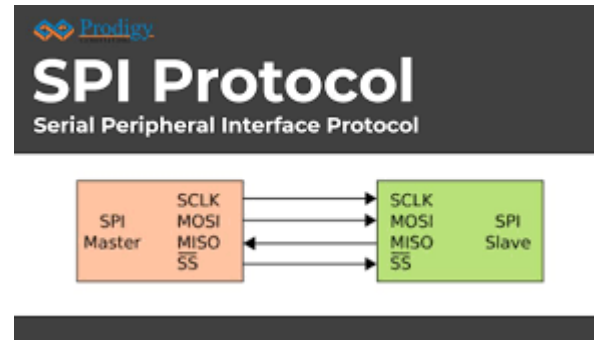
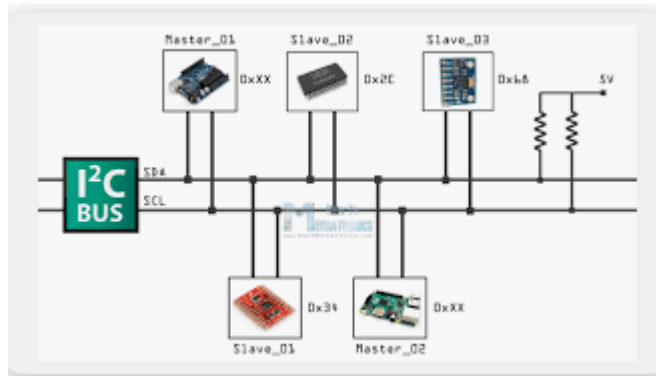
MISO (Master In, Slave Out): This is the data line used by the slave device to send data back to the master device.

MOSI (Master Out, Slave In): This is the data line used by the master device to send data to the slave device.

SCK (Serial Clock): The clock signal, generated by the master, which synchronizes the data transmission.

CS (Chip Select) or SS (Slave Select): A control line that the master uses to enable and disable specific slave devices. Each slave on the bus typically requires a dedicated CS line

USART, I2C & SPI



4. INTERRUPTS:

- PIC microcontrollers have 20 **internal** interrupts and three **external** interrupt sources that are connected to various peripherals such as ADC, USART, Timers, and CCP(A CCP protocol can refer to several things, most commonly the [CAN Calibration Protocol](#) **(CCP)**)

5. I/O PORTS:

- Consider the PIC16 series, which has five ports: Port A, Port B, Port C, Port D, and Port E.
- Port A: It is a 7-bit wide port that can be used for input and output. The TRISA register's status determines whether it is utilised as an input or output port.
- Port B: It's an 8-bit port. This port can be used as both an input and an output. In addition, four of its bits are configurable in input mode based on interrupt signals.
- Port C: It is likewise an 8-bit port that can be used as an input or output port, depending on the TRISC register's status.
- Port D: Unlike Ports A, B, and C, this 8-bit port is not an input/output port; instead, it serves as a slave port for connecting to the CPU. All pins on Port D should have Schmitt Trigger buffers while in I/O mode.
- Port E: It is a 3-bit port that serves as an additional feature of the A/D converter's control signals.

6. CCP MODULE:

- A CCP module can operate in one of three modes:
- Capture Mode: When a signal is received, time is captured, or we can say that when the CCP pin is high, the value of the Timer1 is captured.
- Compare Mode: It functions like an analogue comparator, which means that when the value of timer 1 hits a certain threshold, it generates an output signal.
- PWM Option: This mode provides a configurable 10 bit resolution pulse and duty cycle.

7. TIMERS:

- Timers and counters are useful because they can **keep track of time and count**. Timer0, Timer1, Timer2, and Timer3 are the four timers that a PIC microcontroller can have (depending on the family). Timer0 and Timer2 are 8-bit timers, while Timer1 and Timer3 are 16-bit timers that can be used as counters. These timers operate in accordance with the modes selected.

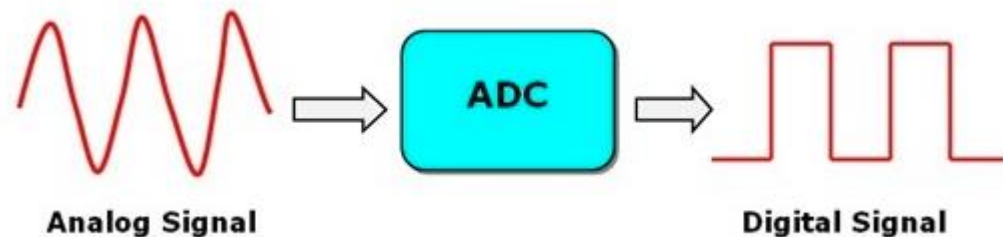
8. D/A CONVERTER:

- The PIC Microcontroller has no analogue outputs. We need to employ an external Digital-to-Analog Converter to get analogue output (DAC). It can convert 8 bits of digital data from the PIC microcontroller's eight digital outputs



9. A/D CONVERTER:

- The PIC Microcontroller has an analogue to digital converter with eight channels and a resolution of ten bits. ADCON0 and ADCON1 are special function registers that govern the A/D converter's operation. The converter's lower bits are stored in the ADRESL register (8 bits), while the upper bits are saved in the ADRESH register. Its operation necessitates a 5V analogue reference voltage.



10. OSCILLATORS:

They are primarily used to generate stable frequency sources for clocks in computers

- **Oscillators** are used to generate timing. External oscillators, such as crystals or RC oscillators, are used in PIC microcontrollers. The crystal is connected between two oscillator pins in crystal oscillators, and the value of the capacitor attached to each pin controls the oscillator's mode of operation. Low-power mode, crystal mode, and high-speed mode are the three modes available. The clock frequency of RC oscillators is determined by the value of the Resistor and Capacitor. The clock frequency is adjustable between 30 kHz and 4 MHz.
- A **Watchdog timer**, which resets the microcontroller in the event of any software failure, and a Brownout reset, which resets the microcontroller in the event of any power fluctuation, are among the other unique peripherals.

- Key Differences Between PIC and ARM:
- Architecture: PIC uses Harvard architecture (separate memory for code and data), while ARM typically uses Modified Harvard or Von Neumann architectures.
- Performance and Speed: ARM processors are significantly faster than PIC microcontrollers, often executing most instructions in a single clock cycle compared to PIC's four cycles.
- Bit Depth: PIC microcontrollers are commonly 8-bit or 16-bit, with some 32-bit (PIC32), while ARM is primarily 32-bit or 64-bit.
- Power Consumption: PIC is generally more power-efficient, making it ideal for battery-operated, simple, or small-scale applications.
- Complexity and Usage: ARM processors are designed for high-performance applications like mobile devices, digital signal processing, and complex automation, whereas PIC is suited for embedded systems, consumer electronics, and basic control tasks.
- Cost: PIC microcontrollers are generally lower in cost and simpler to learn compared to the more advanced ARM processor

Feature 	PIC Microcontroller	ARM Processor
Manufacturer	Microchip	Multiple (e.g., Apple, Samsung, TI)
Bit Width	8, 16, or 32-bit	32 or 64-bit
Architecture	Harvard	Modified Harvard/Von Neumann
Speed	4 clock cycles/instruction	1 clock cycle/instruction
Application	Simple, low-power embedded tasks	High-performance/complex tasks
Cost	Generally lower	Higher